

Cloud-Native Intrusion Detection with Gaussian Anomaly Detection

AI for Cybersecurity — Module 1 Lab

Ruthvik Nath Bandari

Dataset: CIC-IDS-2017 (Canadian Institute for Cybersecurity, UNB)

Colab notebook: `module1_lab_ids_cicids2017.ipynb`

Introduction

This lab implements a network intrusion detection system in Google Colab using the anomaly detection approach of Parisi (2019, ch. 5). Instead of a supervised classifier, the detector estimates the probability density of *benign* traffic and flags flows in low-probability regions. The choice is deliberate: a SOC has unlimited normal traffic but few labelled attacks, and a density model can flag techniques it has never seen. All eight CIC-IDS-2017 capture files were used — 2.83 million flows, fifteen classes.

Methodology

A data-quality audit preceded cleaning; each defect drove one preprocessing step. Headers carried leading spaces; `Fwd Header Length` appeared twice, byte-identical; `Flow Bytes/s` and `Flow Packets/s` held 4,376 infinities and 1,358 NaNs from division by zero-duration flows; eight columns were constant; 329,896 rows were duplicates; and the Web Attack labels contained a corrupt `U+FFFD` byte preventing those families from grouping. Affected rows (0.101%) were dropped rather than imputed: inventing a flow rate never observed is worse than losing a thousandth of the data. Cleaning yielded 2,497,980 flows and 69 features.

Three transformations followed. A signed `log1p` compressed the heavy tails of packet counts and durations, cutting mean absolute skew from 45.3 to 7.3. Features correlating above $\text{Irl} = 0.95$ with a retained feature were pruned, reducing 69 to 40 — necessary because the model inverts the covariance matrix Σ , which collinear features make singular. A quantile transform then mapped features toward normality. All three were fitted on benign training data alone, preventing leakage.

Training received 60% of benign flows and **zero attacks**; validation and test each took 20% of benign flows and half the attacks. A multivariate Gaussian was implemented from first principles, computing $\log p(x)$ by Cholesky factorisation for stability, with ridge regularisation ensuring positive-definiteness. The threshold ε was chosen on validation by maximising F1, since accuracy is useless here — flagging nothing scores 83%.

Results

The single Gaussian reached ROC-AUC 0.921 but a 20.9% false-positive rate, because it assumes benign traffic forms one elliptical cloud. Real traffic is multimodal: DNS lookups, bulk downloads and keep-alives differ. A Gaussian Mixture Model with $k = 10$, selected on validation PR-AUC, lifted ROC-AUC to **0.966** and PR-AUC from 0.787 to **0.930** while halving the false-positive rate to 10.3%. An Isolation Forest benchmark scored worse (ROC-AUC 0.848), confirming the gain came from modelling, not an easy dataset.

Per-family recall exposed a capability boundary aggregate metrics hide. Seven of fourteen families exceeded 90% recall, including all eleven Heartbleed flows — caught by a model that never saw one. But SSH-Patator (9.2%), Bot (26.6%) and all three Web Attack families ($\leq 4.8\%$) were missed. The cause is structural: brute-force and bot traffic look like ordinary sessions *per flow*, suspicious only across many flows, while injection attacks live in HTTP payloads flow statistics never encode. Tightening ε to a 1% false-positive budget gave 95.8% precision at 43.6% recall — the configuration a SOC would deploy.

Current trend: cloud-native IDS

This architecture underpins cloud-native detection. Signature-based IDS assumes a stable perimeter, but containers live for minutes and serverless functions have no host to enrol; a model of behaviour survives that churn where a rule naming an IP does not. The economics fit too: fitting is offline, while scoring is one matrix multiply over kilobytes of state — 2.5 million flows scored in seconds on a free Colab CPU. Amazon GuardDuty and Microsoft Defender for Cloud apply comparable analytics to VPC flow

logs. Because the detector is just a pipeline, a density and a threshold, it can be versioned, reviewed and CI-tested — the **detection-as-code** practice now standard in mature SOC's.

Conclusion

A model trained without a single labelled attack detected most intrusions in held-out traffic, and the decisive insight was diagnostic rather than algorithmic: recognising that benign traffic is multimodal produced a larger gain than any parameter tuning. Equally instructive was learning what the model cannot do — per-family recall turned a good aggregate score into an honest statement of where this layer belongs. Interrogating a result until its limits are explicit is the habit I most want to carry into security engineering work.

Limitations: CIC-IDS-2017 is testbed traffic from 2017, cleaner than production; rare families rest on tiny samples; and the training window is assumed attack-free, which a real deployment must verify.

References

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